Software Testing Assignment

Module 2(Manual Testing)

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1) What is Exploratory Testing?

Exploratory Testing is a type of software testing in which the tester is free to select any possible methodology to test the software and exploratory testing checks the functionality and operations of the software as well as identify the functional and technical faults in it.

2) What is traceability matrix?

A Traceability Matrix is a document that correlates any two baseline documents that require a many-to-many relationship to check the completeness of the relationship. It is used to track the requirements and to check whether the current project requirements are met.

3) What is Boundary value testing?

Boundary value testing is one of the most widely applied black box testing methodologies across the globe. It is defined as a black box test technique that identifies the errors and bugs between the extreme limits or boundaries of a software program or module.

4) What is Equivalence partitioning testing?

In equivalence partitioning, equivalence classes are evaluated for given input conditions. Whenever any input is given, then type of input condition is checked, then for this input conditions, Equivalence class represents or describes set of valid or invalid states.
The Valid partition is bounded by the values 1 and 100 Plus there are 2 Invalid partitions.

5) What is Integration testing?

Integration Testing is a software testing techniques that focuses on verifying the interactions and data exchange between different components or modules of a software application The goal of integration testing is to identify any problems or bugs that arise when different components are combined and interact with each other.

6) What determinesthe level of risk?

likelihood (probability) of a risk event occurring and the magnitude (impact/severity) of the consequences if it does occur. This is often expressed by the formula: Risk = Likelihood × Impact.

7) What is Alpha testing?

Alpha testing is internal, controlled, early-stage testing done by developers or QA teams to catch bugs.

8) What is beta testing?

Beta testing is external, real-world testing done by end-users to validate usability before launch.

9) What is component testing?

Component is a software item that can be tested in separate or in isolation. It means a unit or component is the smallest part of software.
Unit testing done by developer side and the testing of individual software components and it is a level where the software/system are tested.

10) What is functional system testing?

Functional system testing is a quality assurance process where a complete and fully integrated software system is evaluated to ensure it meets its specified functional requirements.

11) What is Non-Functional Testing?

Non-Functional Testing: Testing the attributes of a component or system that do not relate to functionality, e.g. reliability, efficiency, usability, interoperability, maintainability and portability.
Non functional testing performe after functional testing.
1)Performance testing 2)Load testing 3)Migration testing 4)Stress testing 5)Penetration testing 6)Installation testing 7)Security testing 8)Compatibility testing 9)Volume testing

12) What is GUI Testing?

GUI means graphical user interface, gui testing is the process of testing the system's user interface of the System under test. GUI testing involves checking the screens with the controls like buttons, menu, checkbox, icons, and bars like tool bar, menu bar, dialog boxes etc.

13) What is Adhoc testing?

The Error guessing is a technique where the experienced and good testers are encouraged to think of situations in which the software may not be able to cope.
Main aim of this testing is to find defects by random checking. Adhoc testing can be achieved with the testing technique called Error Guessing

14) What is load testing?

Load testing is a performance testing to check system behavior under load. load testing is a kind of performance testing which determines a system's performance under real-life load conditions. This testing helps determine how the application behaves when multiple users access it simultaneously.

15) What is stress Testing?

Stress testing is used to test the stability & reliability of the system. system is stressed beyond its specifications to check how and when it fails. Performed under heavy load like putting large number beyond storage capacity, complex database queries, continuous input to system or database load. Stress testing is also known as endurance testing.

16) What is white box testing and list the types of white box testing?

White box testing is a software testing technique that involves testing the internal structure and workings of a software application. White box testing is also known as structural testing or code based testing, and it is used to test the software's internal logic, flow, and structure.

1)Statement coverage
2)Decision coverage
3)Condition coverage

17) What is black box testing? What are the different black box testing techniques?

In Black-box testing is a type of software testing in which the tester is not concerned with the software's internal knowledge or implementation details but rather focuses on validating the functionality based on the provided specifications or requirements.

Equivalence partitioning
Boundary value analysis
Decision tables
State transition testing
Use-case testing
Other black box testing

18) Mention what are the categories of defects?

Critical (unsafe), Major (unusable/un-saleable), and Minor (workmanship issue)

19) Mention what bigbang testing is?

Integration testing is a type of testing that is used to check the functionality of integrated components of a software system. It is usually performed after unit testing and before validation testing. In integration testing, individual software components are combined and tested as a group. The purpose of this testing is to check if the components work together as expected. Integration testing can be either bottom-up or top-down.

20) What is the purpose of exit criteria?

Exit criteria for STLC phases are actions or tasks that must be finished before the current phase is over and the next one begins. Hence, exit criteria requirements must be fulfilled to end the STLC phase,

the following requirements must be fulfilled for the test cases development phase to be completed when Test cases must be written and examined and test data needs to be prepared and recognized.

21) When should "Regression Testing" be performed?

When application have any major changes and after updates previous application or option are working properly or not is there any damage or not? We have to check this in regression testing.

22) What is 7 key principles? Explain in detail?

1) Testing shows presence of Defects : Testing can reveal bugs, but cannot prove that the software is 100% defect-free.

2) Exhaustive testing is Impossible! : We cannot test everything.

3) Early Testing : Start testing as early as possible.

4) Defect Clustering : Most defects are usually found in a small number of modules.

5) The pesticide Paradox : Update tests regularly.

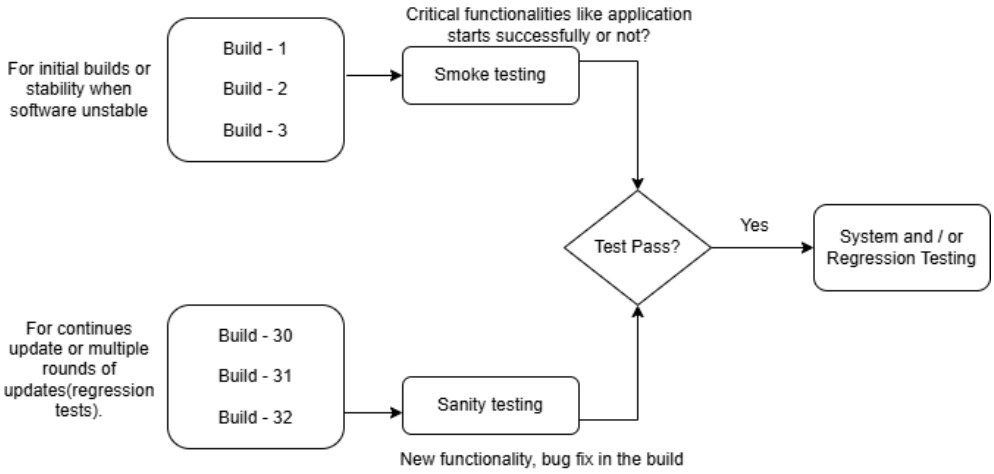
6) Testing is Context Dependent : Different software needs different testing methods (e.g., banking vs gaming).

7) Absence of Error Fallacy : Even if no bugs are found, the product is useless if it doesn't meet requirements.

23) Difference between QA v/s QC v/s Tester

Quality Assurance	Quality Control	Testing
Activities which ensure the implementation of processes, procedures and standards in context verification to of developed software and intended requirements.	Activities which verification software with the ensure developed respect to documented (or not in some cases) requirements.	Activities which ensure identification the of bugs/error/defects in the Software.
Focuses on processes and procedures rather than conducting actual testing on the system.	Focuses on actual testing by executing Software with intend to identify bug/defect through implementation of procedures and process.	Focus on actual testing
Process oriented actvities	Product oriented activities	Product oriented activities
Preventive activities	It is a corrective process	It is a preventive process

24) Difference between Smoke and Sanity?



25) Difference between verification and Validation

Verification	Validation
Verification refers to the set of activities that ensure software correctly implements the specific function	Validation refers to the set of activities that ensure that the software that has been built is traceable to customer requirements.

It includes checking documents, designs, codes, and programs.	It includes testing and validating the actual product.

26) Explain types of Performance testing

Load testing, stress testing, sclability testing, capacity testing, volume testing etc….

27) What is Error, Defect, Bug and failure?

Error : A mistake in coding is called error
Defect : Error found by tester is called defect
Bug : Defect accepted by development team then it is called bug
Failure : Build does not meet the requirements then it is failure

28) Difference between Priority and Severity

Severity measures a defect's technical impact on functionality (e.g., blocker, minor), assessed by QA, while Priority indicates the urgency of fixing a bug based on business value (e.g., high, low), determined by managers/clients.

29) What is Bug Life Cycle?

New – assigned – OPEN – fixed – pending retest – retest – verified – closed

DUBLICATE

REJECTED

DEFFERED

NOT A BUG

Re-opened

30) Explain the difference between Functional testing and Non-Functional testing

Functional Testing	Non-functional Testing
It verifies the operations and actions of an application.	It verifies the behavior of an application.
It is based on requirements of customer.	It is based on expectations of customer.
It helps to enhance the behavior of the application.	It helps to improve the performance of the application.
Functional testing is easy to execute manually.	It is hard to execute non-functional testing manually.

Functional Testing	Non-functional Testing
It tests what the product does.	It describes how the product does.

31) To create HLR & TestCase of

1) (Instagram , Facebook) first page and chat functionality

2) Facebook Login Page : https://www.facebook.com/

https://docs.google.com/spreadsheets/d/19xkxD7hhI0jtTiJz5KGRM0LY4_0CnRGK/edit?usp=drive_link&ouid=108435339942313951523&rtpof=true&sd=true

32) What isthe difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)?

	SDLC	STLC
Focus	Software product development	Product testing
Purpose	Create a great quality software product	Prevent bugs from slipping into the final stages of development
Sequence	Before testing begins	After the SDLC is completed
Requirements	Centered around user's needs and expectations	Centered around the product development
Executors	Business analysts and developers	Testing team
End goal	Deploy a product or new features	Find bugs and defects and report about them to the development team

33) What isthe difference between test scenarios, test cases, and test script?

A test scenario is a high-level description of a functionality or feature to be tested. It focuses on the business process from a user's perspective rather than technical steps.

A test case is a set of specific conditions or steps used to verify a particular feature of an application. It is derived from a test scenario and provides the exact details for a tester to follow manually.

A test script is a set of code or programmed commands used primarily in automated testing to execute test cases automatically.

34) Explain what Test Plan is? What isthe information that should be covered.

A test plan is a comprehensive document—or structured roadmap—that details the strategy, objectives, resources, schedule, and scope of software testing for a project.

35) What is priority?

Priority indicates the urgency of fixing a bug based on business value (e.g., high, low), determined by managers/clients.

36) What is severity?

Severity measures a defect's technical impact on functionality (e.g., blocker, minor), assessed by QA.

37) Bug categories are…

Bug categories in software testing are classifications used to identify, manage, and fix software defects based on their nature, origin, or impact.

38) Advantage of Bugzila .

Bugzilla is a web-based, open-source defect tracking system known for its stability and specialized focus on bug management. Originally developed by the mozilla foundation, it is widely used by major open-source projects, ex Linux, Apache.

39) Difference between priority and severity

Severity measures a defect's technical impact on functionality (e.g., blocker, minor), assessed by QA, while Priority indicates the urgency of fixing a bug based on business value (e.g., high, low), determined by managers/clients.

40) What are the different Methodologiesin Agile Development Model?

Scrum & Kanban

Scrum is the most popular Agile framework, used by nearly 87% of practitioners. It breaks projects down into small, time-boxed iterations called sprints.

Kanban is a visual workflow management methodology focused on continuous delivery and optimizing efficiency.

41) Explain the difference between Authorization and Authentication in Web testing.What are the common problems faced in Web testing?

Authentication (AuthN): The process of verifying a user's identity—confirming they are who they claim to be. It typically involves validating credentials like usernames, passwords, biometrics, OTPs.

Authorization (AuthZ): The process of determining what an authenticated user is allowed to do or see within the system. It manages access rise and permission based on predefined rules or roles

42) To create HLR & TestCase of WebBased (WhatsApp web) 1. WhatsApp Web : https://web.whatsapp.com/

https://docs.google.com/spreadsheets/d/19xkxD7hhI0jtTiJz5KGRM0LY4_0CnRGK/edit?usp=drive_link&ouid=108435339942313951523&rtpof=true&sd=true

43) Create TestCases on Whatsapp Group Chat

https://docs.google.com/spreadsheets/d/19xkxD7hhI0jtTiJz5KGRM0LY4_0CnRGK/edit?usp=drive_link&ouid=108435339942313951523&rtpof=true&sd=true

44) Write a scenario of only Whatsapp chat messages

Verify that the user can set a chat wallpaper.

Verify that the user sets privacy settings like turning on/off last seen, online status, read receipts, etc.

Verify that the user can take the complete chat backup of his chats.

Verify that the user can update the phone number that is used by the WhatsApp application.

Verify that the user can disable/delete his Whatsapp account.

Verify that the user can check data usage by images, audio, video, and documents in WhatsApp chats.

45) Write a Scenario of Pen

Verify the type of pen, whether it is a ballpoint pen, ink pen, or gel pen.

Verify that the user is able to write clearly over different types of papers.

Verify if the pen is with a cap or without a cap.

Verify the color of the ink on the pen.

Check the odor of the pen's ink on writing over a surface.

Verify the pen's performance on different paper textures (smooth, rough, glossy, etc.) to ensure consistent ink dispersion and grip.

Verify the surfaces over which the pen is able to write smoothly apart from paper e.g. cardboard, rubber surface, etc.

Check that the pen's ink should not leak in case it is tilted upside down.

Verify if the pen's ink should not leak at higher altitudes.

Verify if the text written by the pen is erasable or not.

46) Write a Scenario of Pen Stand

Verify the maximum number of pens/pencils the stand can hold without overflowing or losing stability.

Check if the stand remains upright and does not tip over when it is half-full or contains pens of different weights.

Verify if standard-sized pens, markers, and pencils fit easily into the slots or compartment.

Ensure pens can be inserted and removed smoothly without getting stuck or damaging the stand's interior.

47) Write a Scenario of Door

Verify if the door is single door or bi-folded door.

Check if the door opens inwards or outwards.

Verify that the dimension of the doors are as per the specifications.

Verify that the material used in the door body and its parts is as per the specifications.

Verify that color of the door is as specified.

Verify if the door is sliding door or rotating door.

48) Write a Scenario of ATM

Verify the type of ATM machine, if it has a touch screen, both keypad buttons only, or both

Verify that on properly inserting a valid card different banking options appear on the screen

Verify that the touch of the ATM screen is smooth and operational

Verify that the user is presented with the option to choose a language for further operations

Verify that the user is allowed to get account details like available balance

Verify that the user is provided the option to get the transaction details in printed form

Verify that the user’s session timeout is maintained

49) When to used Usablity Testing?

Usability testing should be conducted iteratively throughout the entire product development lifecycle

50) What is the procedure for GUI Testing?

The procedure for Graphical User Interface (GUI) testing involves a systematic verification of an application's visual elements—such as buttons, menus, icons, and layout—to ensure they function correctly and provide a seamless user experience.

51) Write a scenario of Microwave Owen

Verify that the dimensions of the oven are as per the specification provided

Verify that the oven's material is optimal for its use as an oven and as per the specification

Verify that the oven heats the food at the desired temperature properly

Verify the ovens functioning with the maximum attainable temperature

Verify the ovens functioning with minimum attainable temperature

Verify that the oven's door gets closed properly

Verify that the oven's door opens smoothly

Verify that the temperature regulator is smooth to operate

Verify that the temperature regulator works correctly

Verify that the power cord of the oven is long enough

Verify that the usage instruction or user manuals have clear instructions

52) Write a scenario of Coffee vending Machine

Verify that outer body, as well as inner part's material, is as per the specification

Verify that the machine's body color as well brand is correctly visible and as per specification

Verify the input mechanism for coffee ingredients-milk, water, coffee beans/powder, etc

Verify that the quantity of hot water, milk, coffee powder per serving is correct

Verify the power/voltage requirements of the machine

Verify that coffee should not leak when not in operation

Verify the amount of coffee served in single-serving is as per specification

Verify that the digital display displays correct information

53) Write a scenario of chair

Verify that the chair is stable enough to take an average human load

Check the material used in making the chair-wood, plastic etc

Check if the chair's leg are level to the floor

Verify the paint's type and color

Verify if the chair's material is brittle or not

Check if cushion is provided with chair or not

Verify that the weight of the chair is as per the specifications

54) Create Test Cases on Compose Mail Functionality.

https://docs.google.com/spreadsheets/d/1RBwCwWTJgO_UCj8010t311KNLtI-Zwpo/edit?usp=drive_link&ouid=108435339942313951523&rtpof=true&sd=true

55) Online shopping to buy product (flipkart)

https://docs.google.com/spreadsheets/d/1RBwCwWTJgO_UCj8010t311KNLtI-Zwpo/edit?usp=drive_link&ouid=108435339942313951523&rtpof=true&sd=true

56) Write a Scenario of Wrist Watch

Verify the type of watch – analog or digital

Verify the material of the watch and its strap

Verify the dimension of the watch is as per the specification

Verify the weight of the watch

Verify that the numbers in the dial are clearly visible or not

Verify if the brand of the watch and check if its visible in the dial

Verify if the watch comes with any guarantee or warranty

Verify if the dial's glass/plastic is resistant to minor scratches or not

57) Write a Scenario of Lift(Elevator)

Verify the dimensions of the lift

Verify the type of door of the lift is as per the specification

Verify the type of metal used in the lift interior and exterior

Verify the capacity of the lift in terms of the total weight

Verify the buttons in the lift to close and open the door and numbers as per the number of floors

Verify that the lift moves to the particular floor as the button of the floor is clicked

Verify if there is an emergency button to contact officials in case of any mishap

Verify the performance of the floor – the time taken to go to a floor

58) Write a Scenario of Whatsapp payment

1) Scenario: Send money via WhatsApp Pay

Given: User has WhatsApp Pay enabled and UPI linked

When: User opens a chat > Attachment > Payment

And: Enters amount ₹500 and UPI PIN

Then: Payment processed, recipient notified

2) Scenario: Send money with insufficient balance

Given: User has ₹100 in linked UPI account

When: User tries to send ₹500

Then: Transaction fails with 'Insufficient balance' error

3) Scenario: Request money from contact

Given: User has WhatsApp Pay enabled

When: User sends money request of ₹200 to a contact

Then: Contact receives notification with payment request

4) Scenario: View payment history

Given: User has made previous transactions

When: User opens Payments > Transaction history

Then: List of all transactions shown with date, amount, status

5) Scenario: Cancel payment before confirmation

Given: User is on payment confirmation screen

When: User clicks Cancel

Then: Payment cancelled, no deduction made